Progress Report - Rasika Menik Hitihami Mudiyanselage

Application

Thesis title

Thesis title
Time Series Analysis with Features from Ordinal Patterns

Do you need to change your thesis title?
Yes

What is your new thesis title?
Ordinal Pattern Features: Statistical Properties and Their Applications in Time Series Clustering

Progress Report (Self Review)

Is this your first Progress Report?
No

Do you anticipate requiring additional resources such as laboratory or other access, computer, space, over the next six months?
No

Progress to date

Progress to date
The initial simulation results were not successful, requiring the simulations to be repeated using different model classes and modified experimental conditions. Several adjustments were made to the experimental setup in order to improve the separability and stability of the clustering results. After re-running the simulations with the revised model configurations, the analysis produced successful outcomes and demonstrated improved performance under the new experimental setup.

Achievements
During this six-month period, I was unable to achieve any externally recognised accomplishments. However, I continued to make progress in my research activities by refining the experimental setup, improving the simulation framework, and successfully obtaining more reliable results for the study.

Other work
Yes

No, I was able to manage the tutoring responsibilities alongside my other commitments.

Overall assessment
Good

Please give details of any concerns

NA

Are you hoping to submit your thesis in the next 12 months?
No

Review of Goals

Goal	Status	Set by	Deadline
Entropy-Complexity Plane Discrimination of Simulated Time Series Models	Outstanding	Progress Report	30 May 2026
Application of Entropy-Complexity Plane to Real-World Time Series Classification	Outstanding	Progress Report	30 May 2026

Comments on progress of Goals
The original research plan in Phase 1 was revised after the initial simulation experiments did not produce sufficiently clear separation in the entropy–complexity (H-C) plane. The initial ARMA, AR, and MA model configurations were unable to fully visualise the expected discrimination between classes. As a result, the simulation framework was redesigned to incorporate both deterministic and stochastic models together with their convex combinations.

Following this modification, the updated experimental setup produced significantly improved visual separation within the H-C plane. The revised simulations enabled a more detailed investigation of time series clustering behaviour using ordinal-pattern-based entropy and complexity measures. In addition to the visual analysis, clustering performance was evaluated using silhouette-based clustering methods, which demonstrated improved discrimination between model classes.

The clustering structure observed in the entropy–complexity representation was further validated using the DBSCAN algorithm. The DBSCAN results confirmed that the revised feature space preserves meaningful structural differences between the simulated time series models and supports reliable cluster identification.

Phase 2 is yet to complete and will be finished within the next two months.

New Goals

Do you wish to add a Goal?
Yes

Title
Theoretical analysis and improvement of the discrete Fisher Information measure, including derivation of its maximum using Lagrange multipliers.

Type
Research

Fixed deadline
30 Nov 2026

Reminder

Reminder sent

2 weeks before the deadline

Reminder sent to

Candidate

Requirements to complete goal

A clear mathematical derivation of the maximum Fisher Information using Lagrange multipliers.

A written theoretical explanation showing the steps and reasoning.

Numerical verification confirming the analytical result.

Supervisor approval confirming correctness and clarity.

Confirmation of goal completion requires:

Manual confirmation from:

Supervisor

Title

Create a manuscript based on the simulation setup and apply it with the real data

Type

Scholarly engagement

Fixed deadline

30 Nov 2026

Reminder

Reminder sent

2 weeks before the deadline

Reminder sent to

Candidate

Requirements to complete goal

A complete manuscript draft including simulation results and real-data application.

Reproducible analysis files (simulation code, real-data processing, figures).

Validation results showing agreement between simulated and real-data behaviour.

Supervisor approval confirming the manuscript is ready for submission.

Confirmation of goal completion requires:

Manual confirmation from:

Supervisor